Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx}x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

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Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q001 Written

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3, and from 3 to $3 + h$.

Solution

1. Use $[f(b) - f(a)]/(b - a)$. For the h-form, factor $f(3 + h) - f(3) = h(8 + h)$.

Final answer

$4; \quad 8 + h$

Q002 Written

Find the average rate of change for the three source Try functions on their stated intervals.

Solution

1. Evaluate both endpoint values, subtract, and divide by the change in x; cancel h in the variable-endpoint case.

Final answer

$2; \quad -5/3; \quad h - 6$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q003 Written

Find the average rate of change for all nine source Exercise functions, including fixed and h-form intervals.

Solution

1. Keep the endpoint order consistent and simplify before substituting numerical values.

Final answer

Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Solution

1. With $A = \pi d^2/4$, compute $[A(18) - A(10)]/(18 - 10) = 7\pi$.

Final answer

$7\pi\ \text{cm}^2/\text{cm}$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q005 Written

Find the source limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Solution

1. Direct substitution is valid when defined; otherwise factor/cancel first, then put the limiting value into the simplified expression.

Final answer

$-2; \quad 6$

Q006 Written

Find the three source Try limits, including polynomial and h-quotient forms.

Solution

1. Test direct substitution, identify 0/0, simplify, and substitute again.

Final answer

Substitute after removing any common factor; polynomial limits are obtained directly.

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q007 Written

Find all nine source Exercise limits, including x-limits and h-limits.

Solution

1. For a polynomial use continuity; for a quotient, factor before taking the limit.

Final answer

Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 Written

If $\lim_{x\to 3}(x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk.

Solution

1. Set the numerator to zero at 3: $k = 6$. Then cancel $x - 3$, giving $L = 1$, so $Lk = 6$.

Final answer

6

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Solution

1. Use $\lim_{h\to 0}[f(-2+h) - f(-2)]/h = \lim(-12+3h) = -12$.

Final answer

$f'(-2) = -12$

Q010 Written

Find the differential coefficients at the given values for the two source Try functions using the definition.

Solution

1. Expand the shifted polynomial fully before cancelling the common h.

Final answer

Substitute a into $[f(a+h) - f(a)]/h$, cancel h, and let h tend to zero.

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q011 Written

Find the differential coefficients for all six source Exercise functions at their stated values.

Solution

1. The resulting constant is the instantaneous slope at the given input.

Final answer

Use the definition at each specified point and simplify exactly.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Solution

1. The differential coefficient is $s'(t) = 9.8t$; hence $s'(10) = 98$.

Final answer

$98\ \mathrm{m\,s^{-1}}$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q013 **Written**

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Solution

1. Expand $f(x+h) - f(x) = 2xh + h^2 + 3h$, divide by h, and let h tend to zero.

Final answer

$f'(x) = 2x + 3$

Q014 **Written**

Differentiate the three source Try functions using the definition.

Solution

1. Do not use a memorised rule; retain the h-limit structure until the cancellation is visible.

Final answer

Expand each $f(x+h) - f(x)$, cancel h, and take the limit.

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q015 Written

Differentiate all nine source Exercise functions using the definition.

Solution

1. Expand, collect the coefficient of h, divide by h, then set $h = 0$.

Final answer

Apply $f'(x) = \lim_{h \to 0}[f(x+h) - f(x)]/h$ to every printed polynomial.

Q016 Written

A regular hexagon has side length x. Write its area and differentiate it using the definition.

Solution

1. Substitute the area formula into the difference quotient; the derivative of the quadratic factor is $3\sqrt{3}x$.

Final answer

$$A = \frac{3\sqrt{3}}{2}x^2, \quad A'(x) = 3\sqrt{3}\,x$$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at x=2, does $\lim_{x\to 2} f(x)$ exist?

Solution

1. A limit depends on nearby values, not on whether the function is defined at the point itself.

Final answer

Yes; the limit is 5.

Q018 MCQ

The average rate of change from a to b is:

Solution

1. Use the secant slope.

Correct option: B

Final answer

$[f(b) - f(a)]/(b - a)$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Solution

1. Evaluate the two endpoints.

Correct option: C

Final answer

4

Q020 MCQ

The average area-rate for diameter 10 to 18 is:

Solution

1. Use $A = \pi d^2/4$.

Correct option: A

Final answer

7π

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q021 MCQ

The h-form average rate is found by:

Solution

1. Simplify the difference quotient.

Correct option: B

Final answer

cancelling the common h

Q022 MCQ

A direct-substitution limit is valid when:

Solution

1. Use continuity where defined.

Correct option: A

Final answer

the expression is defined

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q023 MCQ

The 0/0 form requires:

Solution

1. Remove the common vanishing factor.

Correct option: A

Final answer

factor/cancel first

Q024 MCQ

In the finite-limit challenge, k equals:

Solution

1. The numerator must vanish at $x = 3$.

Correct option: C

Final answer

6

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q025 MCQ

For a polynomial, the limit equals:

Solution

1. Polynomials are continuous.

Correct option: A

Final answer

its value at the point

Q026 MCQ

The definition of the differential coefficient at a is:

Solution

1. Use the difference quotient.

Correct option: A

Final answer

lim [f(a+h)-f(a)]/h

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:

Solution

1. Expand and cancel h.

Correct option: A

Final answer

-12

Q028 MCQ

The stone's velocity at 10 s is:

Solution

1. Use $s'(t) = 9.8t$.

Correct option: B

Final answer

98

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q029 MCQ

After cancelling h, take the limit as:

Solution

1. The definition uses h approaching zero.

Correct option: A

Final answer

$h \to 0$

Q030 MCQ

The derivative function is:

Solution

1. Associate the coefficient with each x.

Correct option: B

Final answer

the differential coefficient for every x

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:

Solution

1. Expand the difference quotient.

Correct option: B

Final answer

$2x + 3$

Q032 MCQ

The area of a regular hexagon of side x is:

Solution

1. Use the standard hexagon area formula.

Correct option: B

Final answer

$(3\sqrt{3}/2)x^2$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q033 MCQ

A function need not be defined at a for its limit to:

Solution

1. Nearby values determine a limit.

Correct option: A

Final answer

exist

Q034 MCQ

The derivative of a constant function is:

Solution

1. The difference quotient of a constant is zero.

Correct option: A

Final answer

0

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 1 Concept Quiz · MCQ

Find the average rate of change of x^2 from 1 to 3.

Solution

1. $[9-1]/2 = 4$.

Correct option: A

Final answer

4

Quiz MCQ 2 Concept Quiz · MCQ

Find $\lim_{x\to 2}(x^2+1)$.

Solution

1. Direct substitution gives 5.

Correct option: B

Final answer

5

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 3 **Concept Quiz · MCQ**

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Solution

1. Expand $(x + h)^2 - x^2$, cancel h, and let h tend to zero.

Correct option: C

Final answer

$2x$

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s(t) = 4.9t^2$, find the velocity at t=5.

Solution

1. $s'(t) = 9.8t$.

Correct option: D

Final answer

49

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the average rate of change of $3x + 1$ from 2 to 7.

Solution

1. A linear function has constant slope 3.

Final answer

3

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\lim_{h\to 0}(6 + 2h)$.

Solution

1. Substitute $h = 0$.

Final answer

6

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the differential coefficient of $2x^2$ at x=3 from the definition.

Solution

1. The difference quotient simplifies to $4x + 2h$, giving 12.

Final answer

12

Quiz WRITTEN 8 **Concept Quiz · Written**

State $\lim_{x \to 2} f(x)$ when f(x)=5 for all x except x=2 where it is undefined.

Solution

1. Nearby values are constantly 5.

Final answer

5

Answer key

Use the question number shown on each page.

Q001 4; $8+h$

Q002 2; $-5/3$; $h-6$

Q003 Apply $[f(b)-f(a)]/(b-a)$ to each printed item exactly.

Q004 $7\pi\ \mathrm{cm}^2/\mathrm{cm}$

Q005 -2; 6

Q006 Substitute after removing any common factor; polynomial limits are obtained directly.

Q007 Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 6

Q009 $f'(-2)=-12$

Q010 Substitute a into $[f(a+h)-f(a)]/h$, cancel h, and let h tend to zero.

Q011 Use the definition at each specified point and simplify exactly.

Q012 $98\ \mathrm{m\,s^{-1}}$

Q013 $f'(x)=2x+3$

Q014 Expand each $f(x+h)-f(x)$, cancel h, and take the limit.

Q015 Apply $f'(x)=\lim_{h\to0}[f(x+h)-f(x)]/h$ to every printed polynomial.

Q016 $A=\frac{3\sqrt{3}}{2}x^2$, $A'(x)=3\sqrt{3}\,x$

Q017 Yes; the limit is 5.

Q018 B

Q019 C

Q020 A

Q021 B

Q022 A

Q023 A

Q024 C

Q025 A

Q026 A

Q027 A

Q028 B

Q029 A

Q030 B

Q031 B

Q032 B

Q033 A

Q034 A

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 D

Quiz WRITTEN 5 3

Quiz WRITTEN 6 6

Quiz WRITTEN 7 12

Quiz WRITTEN 8 5